Komal Saraf

Kolkata 700048 | +91 7439766518
komalsaraf04@gmail.com LinkedIn Github

EDUCATION:

Bachelor of Technology (B.Tech.) - Computer Science *Sep 2022 - June 2026*
Vellore Institute of Technology (B.TECH CSE)
Cumulative GPA - 7.48

SKILLS:

- **Languages**: Python, C, C++, Java, MySQL.
- **Core Competencies**: Problem-solving, Data Structure and Algorithms, Object-oriented Programming, Database and Management System, (SDLC, STLC, test cases, bug lifecycle, Postman, Selenium, Jira, Machine Learning (TensorFlow, Scikit, Matplotlib, PyTorch, Pandas, NumPy), Data Analysis.

PROJECTS:

- **CivicHelp AI** *Dec 2025 – Feb 2026*
  - Architected and deployed a multimodal GenAI pipeline using Vision Transformers and LLMs to process 10K+ civic complaints, achieving 92% classification F1-score and reducing manual triage time by 60%.
  - Built an embedding with vector search system for similarity detection and hotspot clustering, enabling automated repeated-issue detection and improving routing precision by 45%.
  - Implemented RAG-based policy grounding using municipal embeddings to ensure factually aligned outputs, reducing misinformation risk and improving action recommendation reliability by 30%.
  - Predicted rising civic trends and urgency via integrated detection tools, streamlining analysis and improving response efficiency by 35%.
  - Tech Stack: Python, Generative AI, Vision Transformers, Sentence Transformers, RAG, Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib.

- **Heart disease detection** *May 2025- Sept 2025*
  - Developed a predictive analytics system for early detection of heart disease by training a dataset of patient's physical and medical characteristics.
  - Integrated and optimized various machine learning algorithms (e.g., Logistic Regression, Support Vector Machines, Random Forest) to enhance prediction accuracy and efficiency.
  - Attained a prediction accuracy of more than 85% by analyzing 1000+ dataset entries and built visual reports and performance dashboards to communicate insights
  - Skills Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Machine Learning.

- **Desktop Assistant** *Nov 2024- Feb 2025*
  - Developed an automated workflow management system to handle routine operational tasks, resulting in a 20% increase in process efficiency.
  - Designed modular workflows to integrate speech recognition for sub-second voice-to-text conversion and a GUI (Tkinter) for task execution features demonstrating hands-on experience in automation, process streamlining, and user-centric design.
  - Skills used: Automation, OpenCV, PyTorch, Tkinter Python libraries (SpeechRecognition, pyttsx3)

CO-CURRICULARS:

- Python gold badge , Hackerrank.
- C++ gold badge , Hackerrank.
- Member, AI Club, VIT Bhopal 2022 – Coordinated and organized multiple events, workshops, and functions within the AI Club, collaborating with cross-functional teams contributing to community building, knowledge sharing, and enhancing student participation in Artificial Intelligence initiatives.